

A systematic review of AI for predicting glaucoma progression: challenges and recommendations towards clinical implementation

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**A Systematic Review of AI for Predicting Glaucoma Progression: Challenges and Recommendations
Towards Clinical Implementation**

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Abstract

Glaucoma is the leading cause of irreversible blindness worldwide with heterogeneous progression rates. Artificial Intelligence (AI) may enable accurate progression predictions in clinical practice. We conducted a systematic review to survey quantitative AI performance and examine strengths and shortfalls in current AI approaches with future clinical implementation in mind. Two reviewers independently screened studies in English from MEDLINE, Embase, Web of Science, Cochrane CENTRAL and arXiv since 2014 and performed risk of bias assessment on eligible studies using QUADAS-2. 46 reports of 43 unique studies demonstrated moderate to good performance in predicting glaucoma conversion, biological deterioration and progression to surgery. Several challenges for clinical translation remain, including inconsistent reporting, limitations and heterogeneity in study design and poor AI generalisability and transparency. We encourage future studies to adopt robust study design and transparent reporting and propose the first glaucoma-specific list of recommended practices and reporting items for future clinical implementation.

Introduction

Although the age standardised prevalence of glaucoma worldwide has declined over the past three decades, the total number of diagnosed individuals and cases of blindness due to glaucoma continue to rise.¹ Glaucoma remains the leading cause of irreversible blindness worldwide, with an estimated disease burden of over 111 million people in 2040.² Several challenges complicate the effective management of glaucoma, including complex and debated aetiologies, unpredictable disease trajectories and heterogeneous responses to treatments. The natural history of glaucoma is highly variable – recent estimates suggest that one in eight eyes with glaucoma undergoing routine treatment will exhibit rapid progression beyond -1.0 dB per year.³ Early and accurate prediction of progression is critical to the maximal preservation of vision, as it could guide more intensive monitoring and treatment for high-risk patients.

Previous studies have explored associations between diverse risk factors with glaucoma progression, such as higher cup-to-disc-ratios, poorer baseline visual acuity, genetic variants for optic nerve vulnerability, systemic arterial stiffness and cardiovascular diseases.⁴⁻⁷ However, these factors often lack the necessary predictive granularity and as a result, there is currently no clinically applicable tool to guide the personalised treatment of glaucoma based on forecasted risk of progression.

Artificial intelligence (AI) is a broad term which describes technologies that learn and utilise patterns from input data for complex problem solving and may be broadly categorised into machine learning (ML) and deep learning (DL) approaches. ML typically uses structured or engineered features with simpler model designs, whilst DL leverages complex model architectures with hidden layers to extract implicit features from a wide range of inputs. AI has demonstrated unique advantages in tasks such as image processing and time series predictions compared to previous state-of-the-art approaches, which has led to its application in multiple specialised domains such as wildfire detection and global weather forecasting.^{8,9} Similarly, AI's ability to extract latent features from

noisy, high-dimensional data has also made it an attractive and competitive solution for various clinical tasks.¹⁰

Great strides have been made in developing AI models for a range of diagnostic tasks in ophthalmology, such as detection of diabetic retinopathy (DR), glaucoma and age-related macular degeneration (AMD).¹¹⁻¹³ More recently, researchers have explored the utility of AI models in more complex clinical tasks such as predicting glaucoma progression, which presents unique challenges compared to the diagnostic task due to the heterogeneous definitions and reference standards for glaucoma progression and the inherent difficulties of future event predictions based on current or previous data. Nevertheless, many have made progress in accurately predicting glaucoma progression with AI, with some approaching the performance of models for diagnostic tasks.¹⁴⁻¹⁶ AI tools designed for more straightforward clinical tasks such as diabetic retinopathy detection have received FDA approval after demonstrating good performance and safety in real-world settings, giving hope to AI tools for more complex clinical tasks such as glaucoma progression prediction, provided that they also undergo rigorous validation and regulatory assessment.¹⁷ Similar to exploration of AI applications in other fields, AI studies for glaucoma predictions have seen significant growth in research attention and output in recent years, albeit with high heterogeneity and inconsistency in their approaches. It is therefore of interest to researchers, clinicians and developers to systematically map out the current landscape of glaucoma progression forecasting with AI in order to understand the heterogeneous design and outcomes in current approaches and identify key challenges and strategies towards clinical implementation. Although previous analyses explored major trends and limitations in this field, several key issues remain poorly addressed, such as interpreting AI model outputs, integrating AI into existing clinical pathways and consistent reporting items for future studies.¹⁸ Our review placed particular emphasis on these topics which are critical to the robust evaluation and safe implementation of AI tools.

This systematic review aimed to survey the performance of AI models designed to predict glaucoma progression in glaucoma, suspect and ocular hypertension patients. Through a qualitative synthesis of AI performance, we assessed the current predictive accuracy of AI models which may serve as benchmarks for future studies in this field. Additionally, we sought to critically examine the strengths and shortfalls of current AI approaches in diverse domains such as transparency, accessibility, system integration and consistent reporting guidelines with future clinical implementation in mind. Incorporating our findings across various domains of glaucoma AI studies, we provide pragmatic recommendations to facilitate future clinical implementation.

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Results

1. Study selection

4138 records were returned from our searches in Embase (1945), Web of Science (1373), MEDLINE (680), arXiv (110) and Cochrane CENTRAL (30), of which 2767 were unique entries. 129 full-text studies were assessed for eligibility after the title and abstract screen, and 46 reports¹⁹⁻⁶⁴ of 43 unique study cohorts were deemed eligible for qualitative synthesis (Figure 1). Overlapping reports were grouped by study cohorts to highlight potential correlations between their reported outcomes. The full search strategies for each database and the completed PRISMA-DTA checklist were included as supplementary material sections 2 and 3, respectively.

2. Study characteristics

The 43 unique study cohorts collectively included more than 202,207 subjects (two studies did not report number of subjects) from six countries with the vast majority from the United States (28 studies),^{20-24, 27-30, 32, 33, 37-40, 44-46, 48-53, 55, 56, 59, 61-64} followed by South Korea (8),^{26, 34, 35, 41-43, 47, 60} Japan (3),^{19, 25, 58} China (2),^{36, 54} Egypt (1)⁵⁷ and Lithuania (1).³¹ The country of origin was defined here as the origin country of the study cohort. The largest study^{44, 45} drew upon a multi-centre cohort of 39,090 patients, whereas the median cohort size was 1231 subjects. 36 (78%) of the 46 articles were published in the last four years. All 43 studies were retrospective, longitudinal cohort studies. Regarding the recruited population, 25 studies^{19-22, 25, 28, 29, 31-33, 35, 37, 39, 40, 43-45, 47-52, 55, 58, 60, 64} focused on patients with diagnosed glaucoma, 9 studies^{24, 26, 30, 36, 54, 56, 57, 61, 62} recruited patients with ocular hypertension (OHT) or glaucoma suspects, and 9 studies^{23, 27, 34, 41, 42, 46, 53, 59, 63} included both glaucoma and OHT/suspects. 34 studies reported cohort characteristics such as mean age and gender ratios. Importantly, 17 studies^{20-22, 24, 25, 28, 30, 32, 33, 37-40, 44, 45, 50, 52, 55, 62, 63} reported ethnicity or race composition of their cohorts. 34 studies^{19, 22-36, 38-53, 55, 57, 59, 60, 62} reported baseline or overall glaucoma severity with intraocular pressure (IOP), mean deviation (MD) or pattern standard deviation (PSD). Details of the study and cohort characteristics were listed in Table 2.

3. AI architectures, predictive tasks, reference standards and outcome measures

A variety of AI model designs, input modalities, predictive tasks, reference standards and outcome measures were reported in eligible studies. ML models and DL models were deployed by 15^{5, 20-22, 24-26, 33, 40, 44, 45, 48, 50, 55, 57, 60} and 35 studies,^{19-21, 23, 27-32, 34, 36-39, 41-44, 46-48, 50-54, 56-59, 61-64} respectively. Four main types of DL architectures were used, with the most popular architecture being convolutional neural networks (CNN) in 15 studies,^{19, 23, 32, 36-39, 46, 47, 52, 53, 58, 61, 62, 64} followed by feed-forward neural networks (FNN) in 7 studies,^{20, 21, 44, 48, 50, 57, 59} transformers in 7 studies,^{27-30, 51, 54, 63} recurrent neural networks (RNN) in 4 studies^{34, 41, 42, 56} and hybrid recurrent-convolutional neural network (RNN-CNN) in 3 studies.^{31, 43, 56} The majority (34) of studies designed models that receive single modality inputs, namely, inputs that have a singular consistent format, such as a structured vector or a fundus photograph. Notably, 12 studies^{19, 22, 26, 27, 31, 32, 37, 38, 43, 46, 51, 52} developed multimodal models capable of incorporating high dimensional information from different data formats to make predictions.

AI models were trained for three main types of predictive tasks: binary classification, numeric regression and survival prediction tasks. 26 studies^{20-22, 26-33, 36-40, 44, 45, 47-49, 51, 52, 54, 57, 60-62, 64} developed binary classification models, which produce a binary prediction about the progression status of a patient. Depending on the underlying patient cohort, binary classification models either predicted conversion to glaucoma for OHT and suspect patients,^{26, 30, 54, 57, 61, 62} or progression after diagnosis for cohorts with established glaucoma diagnoses. For binary progression predictions in diagnosed glaucoma patients, studies adopted either biologically grounded criteria,^{27, 29-31, 33, 35-40, 47, 48, 61, 64} including event or trend-based visual field analysis or fundus photograph signs of glaucomatous optic neuropathy, or clinical event-based criteria such as escalation of treatment or records of incisional surgery.

Quantitative outcomes reported for the binary classification models are similar to those in DTA studies, including AUC, sensitivity and specificity. Overall ranges of quantitative outcomes, stratified by predictive task and reported outcomes, were outlined in Table 3. The model-level sensitivities and

specificities for models that predicted conversion from OHT/suspect to glaucoma were summarised in Figure 2A, with AUCs ranging between 0.62 and 0.99. Models that predicted progression of established glaucoma cases under biologically grounded and clinical event-based criteria were showcased in Figure 2B and 2C, with AUCs ranges of 0.68-0.93 and 0.65-0.99, respectively. More details on the design and performance of binary classification models were included in Supplementary Table 1.

15 studies^{19, 23-25, 34, 35, 41-43, 46, 53, 55, 58, 59, 63} developed and tested models on the numeric regression task by training models to make quantitative predictions of future functional or structural parameters and comparing to ground truth values from clinical records. These parameters were selected due to their significance in monitoring glaucoma severity and progression, including VF pointwise deviations, VF global indices (MD, PSD, VFI), IOP and OCT indices (RNFL thickness). Similar to binary classification models, numeric regression models were also trained on both OHT/suspect and established glaucoma cohorts, however, numeric regression models were trained to generate the same predictions about functional or structural parameters regardless of the underlying cohort.

Quantitative outcomes of the numeric prediction task include mean absolute errors (MAE) and root mean square errors (RMSE), which report the errors between predicted values and true values. They are not equivalent measures of error, however, as RMSE penalises larger predictive errors. The MAE and RMSE metrics of numeric regression models were summarised in Figure 3, stratified by studies and the predicted parameter, including VF pointwise deviations (Figure 3A), IOP (Figure 3B), MD (Figure 3C), PSD (Figure 3D), VFI (Figure 3E) and RNFL parameters (Figure 3F). Overall, MAE ranges were 1.77-4.77 dB for VF pointwise deviations, 1.06-3.12 dB for MD and 0.74-1.93 dB for PSD and 2.39-11.08% for VFI. RMSE ranges were 2.36-6.13 dB for VF pointwise deviations, 2.1-4.2 mmHg for IOP, 1.0-2.71 dB for MD and 0.4-2.5 dB for PSD. More details on the design and performance of AI models predicting numeric progression-related parameters were included in Supplementary Table 2.

Interestingly, two studies^{26, 50, 56} developed survival models which predicted the risk of progression to glaucoma surgery or conversion from OHT to glaucoma. These models were evaluated at different time marks using the C-index, which measures the rank correlation between the predicted risk scores and the observed time to event. More details on the design and performance of AI survival models were included in Supplementary Table 3.

Although most models performed model evaluation and reported quantitative outcomes on appropriate test subsets unseen during training, only five studies explicitly reported the use of external test sets distinct from the training dataset.^{22, 36, 38, 44, 45}

4. Risk of bias and applicability

The risk of bias and applicability ratings for each domain of QUADAS-2 were reported in Figure 4. Only 18^{20, 22-24, 26-30, 32, 35, 43-45, 47, 48, 50-53, 55} (42%) of the 43 studies received low risk of bias and low applicability concerns across all domains, and seven^{25, 31, 49, 57, 60, 63, 64} (16%) studies received at least one “high” rating across all domains. Studies most frequently showed lapses in the patient selection and flow and timing domains.

5. Secondary outcomes

We reviewed validation strategies, open-source status of the AI model codes and data as well as reporting of explainability features in all included studies, as these factors may have significant impacts on the generalisability and transparency of reported performance and applicability in diverse populations. Seven studies^{21, 22, 36, 38, 43-45} explicitly validated AI models on external datasets. Nine^{21, 23, 28, 37, 43, 53, 56, 61, 63} (21%) of the 43 studies reported open-source access to their AI codes, whilst only two^{21, 37} (5%) studies reported open-source access to their training and testing dataset. 18^{20-22, 26, 28, 29, 32, 35, 36, 39, 40, 44, 46, 48, 50-52, 60, 61, 64} (42%) studies reported using explainability features such as feature importance ranking and saliency maps. Details on the reporting of open-source access and explainability were included in Supplementary Table 4.

Discussion

In this systematic review, we examined a diverse range of studies that developed AI models to predict glaucoma progression using a variety of clinically relevant definitions in OHT, suspect and diagnosed glaucoma patient cohorts. Innovations were noted in the re-formulation of the glaucoma prediction task as binary classification, numeric regression and survival prediction tasks as well as in the design of advanced AI models with novel architectures and hybrid structures. Many studies reported promising predictive performance on internal test datasets considering the complexity of the glaucoma progression prediction task, with a few^{22, 36, 38, 44, 45} demonstrating moderate to good transferability and generalisation on external test data.

The diverse, innovative AI approaches in included studies also came with a high level of between-study heterogeneity, however, as we also noted markedly inconsistent reporting of cohort characteristics, study designs, AI configurations as well as performance outcomes, which poses significant challenges to the robust and systematic evaluation of these models and limits their potentials for future clinical implementation. Furthermore, our exploration of secondary outcomes centred around the safety and fairness aspects of AI models also revealed frequent lapses in these key domains, including poor AI transparency and accessibility, high occurrence of risk of bias and lack of clinical perspectives and guides for the appropriate interpretation and integration of AI outputs. In light of our observations about the strengths and limitations of current AI approaches as well as related research about AI translation in other fields, we proposed recommended good practices and reporting items for future studies to facilitate future clinical validation and implementation.

The clinical implementation of AI tools depends on them demonstrating satisfactory performance in the first instance, although we emphasize that quantitative performance alone is not a sufficient condition for successful clinical adoption. Overall, AI models demonstrated moderate to good predictive performance across all types of predictive tasks. Binary classification models predicting conversion from OHT/suspect to glaucoma achieved a typical sensitivity between 0.56 to 0.97 and a

typical specificity between 0.59 and 0.98 (Figure 2A). Of note, Huang et al.³⁰ were among the first to evaluate the zero-shot glaucoma predictive performance of a commercial large language model (LLM), ChatGPT 4.0, which achieved a sensitivity and specificity of 0.56 and 0.78 respectively at predicting conversion of OHT to glaucoma at 1 year using baseline demographic, clinical, ocular and VF inputs.

Binary classification models predicting progression in diagnosed glaucoma patients adopted different definitions of progression, including biologically grounded definitions such as event- and trend-based criteria using visual field tests or clinical records of treatment escalation such as progression to glaucoma surgery. Progression definition is likely a key factor impacting AI performance. Shon et al.⁴⁷ used a combination of event and trend-based criteria to define progression and reported relatively low sensitivity figures between 0.32 to 0.42 but high specificity figures of 0.95. Conversely, Mohammadzadeh et al.³⁹ adopted a trend-based progression criterion and reported markedly higher sensitivity figures between 0.83 and 0.94, but with lower specificity figures around 0.78. More stringent progression criteria may enable a clearer separation between positive and negative labelled data, which allows models to be more specific or selective in their predictions. The strict criteria may exacerbate class imbalance with fewer positive labels, however, which may negatively impact model sensitivity. We encourage future studies to report clearly the progression criteria they selected and test multiple clinically relevant progression definitions whenever possible.

Larger studies tend to report lower performance outcomes, particularly for binary prediction studies with biologically grounded progression criteria (Figure 2B). Despite having the same trend-based progression criteria (MD slope < -1 dB/year), Schuldiner et al. reported sensitivity and specificity ranging from 0.54-0.67 and 0.53-0.68 respectively on a test cohort of 4585 eyes, whereas Mohammadzadeh et al. reported notably higher sensitivity and specificity ranges from 0.83-0.94 and 0.78-0.79 respectively on a significantly smaller test cohort of around 392 patients. Although the more recent and advanced model design and information-dense input data chosen by

Mohammadzadeh et al. may have been the main driver behind this marked improvement in performance over a similar predictive task, the risk of overfitting on smaller datasets with complex AI models must never be overlooked. Publication bias may also be a partial contributing factor, as published AI studies are generally expected to outperform previous baselines. Potential causes of performance disparities should be more thoroughly evaluated with external testing and more transparent reporting of training and validation accuracy and losses.

Binary classification models that utilise multimodal inputs (combining data from multiple formats, such as images, text and structured vectors) generally demonstrated superior performance at predicting progression to surgery or other clinical record-based events compared to those that processed single modality inputs (Figure 2C). Notably, Wang et al.⁵¹ developed a novel deep learning pipeline to identify eyes at risk of surgery at set time horizons, where spatially preserved VF and OCT data were first concatenated into 3-channel images and converted to feature vectors by a specialised vision transformer (ViT) model. The spatial feature vector could then be merged with structured inputs containing VF, OCT, clinical and demographic data and fed into a fully connected neural network classifier that outputs risk to surgery. The multimodal pipeline achieved sensitivity and specificity ranges of 0.75-0.89 and 0.63-0.93 respectively, for predictions of surgical events up to 5 years.⁵¹ Previous literature⁶⁵ demonstrated that AI models provably perform better with multimodal inputs, as they could learn more optimally when given complementary data in different modalities that could be mapped onto a single latent representation space for the target task. Given the variety of ancillary testing and novel imaging techniques in ophthalmology, developers of glaucoma prediction models and other eye disease models are encouraged to leverage the multimodal nature of ophthalmic data to bolster model performance.

Due to the diverse glaucoma parameters predicted by numeric regression models (Figure 3), including VF pointwise sensitivities, IOP, MD, PSD, VFI and RNFL thickness, direct comparison of performance between studies reporting different outcomes is of little value. We noted that most

studies predicting VF pointwise deviations reported predictive errors between 2 and 6 dB, whereas studies predicting MD, PSD and IOP had tighter error intervals between 1-3 dB, 0.5-2 dB and 2-4 mmHg, respectively (Figure 3B, Figure 3C, Figure 3D). We observed meaningful within-study trends – numeric predictions produce larger errors for longer prediction timelines and more severe glaucoma cases. Garcia et al.²⁴ explored a novel Kalman Filtering (KF) approach for predicting future glaucoma parameters including MD, PSD and IOP. Predictive errors (RMSE) increased from 1.44 dB, 0.73 dB and 2.33 mmHg for MD, PSD and IOP respectively at 1 year to 1.72 dB, 1.04 dB and 3.42 mmHg at 5 years (Figure 3B, Figure 3C, Figure 3D). Park et al.⁴¹ developed a recurrent neural network with novel bidirectional gated recurrent units to predict MD, PSD, VFI and pointwise deviations of 24-2 VFs. Large variations in predictive errors (MAE) were reported for different glaucoma severities. For instance, MD predictions had MAEs of 1.06 dB in early glaucoma cases and 3.12 dB in severe glaucoma cases.⁴¹ We recommend future numeric regression studies to report baseline and overall severity statistics and standard prediction timelines (for example, in yearly or half-yearly steps) to facilitate meaningful comparisons between studies and evaluation for specific use cases, such as near term versus long term projections.

Interestingly, two studies approached the glaucoma progression prediction task as a survival analysis task and built survival models that predicts the probability of progression at different time points. Zhou et al.⁵⁶ built deep learning survival models to predict risk of conversion from OHT to glaucoma, which achieved C-indices ranging from 0.54-0.73. Conversely, Tao et al.⁵⁰ developed traditional and deep survival models that predicted risk of progression to glaucoma surgery, all of which achieved C-indices around 0.77. Although survival modelling of glaucoma conversion and progression showed moderate quantitative performance, they may offer unique advantages in leveraging longitudinal data with inconsistent follow-up or continual updates. Survival models natively handle censored data better compared to binary classification models with cross-sectional inputs and may provide more reliable risk stratifications in common clinical use cases such as ongoing progression risk monitoring.

We encourage more in-depth explorations of survival analysis for predicting glaucoma progression risks for future studies.

Beyond factors that directly contribute to quantitative AI performance, several key limitations in the current AI approaches were noted which pose significant challenges to future clinical adoption. The lack of reporting on clinically relevant and context-dependent performance metrics emerged as one of the significant shortfalls shared by most included studies. Although most studies reported conventional performance metrics such as AUC, sensitivity, specificity, MAE, RMSE and C-indices, their significantly heterogeneous patient cohorts, progression definitions and prediction tasks make the appropriate interpretation of model outputs difficult for real-world clinical scenarios. For instance, Herbert et al.²⁷ developed deep learning models for predicting rapid glaucoma progression defined as MD deterioration worse than -1 dB/year and reported the optimal sensitivity and specificity to be 0.95 and 0.74 respectively with an AUC of 0.87, which indicate good predictive performance overall. The prevalence of rapid progressors is low in their training dataset and in the general glaucoma patient population, however, which could meaningfully impact the clinically relevant interpretation of these performance figures. Using a rapid progressor prevalence of 5.8% reported by Herbert et al.,²⁷ we could calculate the positive and negative predictive values (PPV, NPV) for this model to be 18.4% and 99.6% respectively, suggesting that the model may report many false positives when used in a similar clinical cohort. Although this model may still offer clinical utility for risk screening scenarios owing to its excellent NPV, it may not be appropriate for diagnostic or stratifying pathways where false positives may come with risk of harm or significant costs. We highlight the importance of context-dependent interpretation of performance metrics and recommend future studies to provide clinically relevant performance metrics like NPV and PPV that account for disease prevalence and clinical use cases.

Similarly, numeric regression models are inevitably affected by the statistical uncertainty and errors in its predictions, and the reporting of numeric predictive errors such as RMSE and MAE do not

directly contextualize the model's performance in clinical use cases. Garcia et al.²⁴ innovatively applied Kalman Filtering to predict future MD values in OHT patients, with a reported RMSE of 1.64 dB at two years. Assuming an unbiased model with symmetrical, normal error distributions, the model will have to predict an MD deterioration of at least -4.7 dB at 2 years for a 95% probability of a true underlying MD deterioration of just -2 dB, representing an MD slope of -1 dB/year that may trigger rapid progression pathways in clinical practice. As such, we recommend future numeric regression studies to thoroughly investigate the distribution of predictive errors alongside simple RMSE and MAE metrics to enable output with statistical confidence for better clinical interpretability.

Another key challenge facing most clinical predictive models is the confounding effects of treatment.⁶⁶ In the context of glaucoma predictive modelling, patients may have received a wide variety of interventions prior to or during the prediction period, including topical and systemic medications as well as surgery for glaucoma and other concomitant ocular conditions such as cataract. Individual patients likely also have heterogeneous treatment responses that could be associated with complex patient, institution and regional-level factors. All included studies in this review trained and evaluated models on retrospective data, however, details about treatment history and treatment outcome were unreported and likely absent in most included databases.

Furthermore, prospective investigations may also be limited in their examination of treatment effects due to the ethical restrictions that prevent the withholding of standard of care treatments.

Therefore, current AI modelling approaches in glaucoma and other clinical fields are often limited by a shared, implicit assumption of either no treatment or homogenous treatment profile and responses in their target cohorts, which expose them to a high risk of poor generalisability for patients with different treatment profiles or at different stages of their glaucoma management.⁶⁶

With the practical and ethical limitations of treatment data availability in mind, we recommend that future studies explore novel AI techniques that enable causal inference with treatment effect and treatment response as latent variables.⁶⁷ An example of an innovative AI approach that may prove helpful is the Causal Effect Variational Auto-Encoder (CEVAE), which considers a latent disease state

to be reflected by indirect, proxy variables and influence treatment and outcome.⁶⁸ Future studies are encouraged to explore proxy variable approaches like CEVAE, where a latent glaucoma disease state may be inferred by fitting a model on proxy features such as retinal measurements and treatment-outcome data obtained from data-mining of electronic health records (EHR) and accessing treatment related variables such as IOP, MD, treating institution (and any associated standard of care guidelines) as well as self-reported medication and surgical history.

We investigated several secondary outcomes, including model validation strategies, risk of bias, open-source status, and explainability features to evaluate the generalisability, safety and transparency of current AI approaches, all of which are critical considerations for future clinical implementation. We would like to highlight prevalent shortfalls in these domains, including lack of external validation, risk of bias in study design, poor transparency in data and AI methods, limited use of explainability features and poor reporting on race and ethnicity statistics. Due to heterogeneity in patient cohort, progression criteria and other aforementioned confounding factors such as latent treatment effects, AI models may have poor generalisability when facing “domain shift”, namely, a non-trivial change in the underlying data distribution. External validation on diverse datasets is a powerful test of AI generalisability against domain shift, however, only seven studies^{21, 22, 36, 38, 43-45} reported performance outcomes from external validation datasets. Notably, Baxter et al.²¹ evaluated a single-centre trained AI model predicting the need for glaucoma surgery on a large, external dataset which returned a markedly lower AUC of 0.49 compared to an AUC of 0.67 from internal validation. We therefore recommend all future studies to perform external, ideally multicentre validation wherever possible.

In addition to improper validation strategies, risk of bias may also pose significant challenges to the generalisability of AI models. We found that over half of the included studies carried an unclear or high risk of bias or applicability according to QUADAS-2 criteria (Figure 4). Most lapses were found in the patient selection domain, where studies often adopt non-consecutive or non-randomised

recruitment strategies such as case control or convenience sampling and report limited cohort characteristics. Similarly, several studies were graded as unclear in the flow and timing domain, as some subjects did not reach final analysis due to incomplete data and handling of missing data was not well reported. Future studies are recommended to adopt robust patient selection strategies such as consecutive or randomised recruitment and to report comprehensive cohort characteristics, including characteristics of subgroups, to facilitate thorough assessment of the generalisability of study findings.⁶⁹

Transparency is another crucial consideration towards the integration of AI models in clinical workflows. Modern AI models are data-driven, “black box” systems that are susceptible to biases within its training data, making transparency in data and model design critical for fair evaluation and trust-building amongst developers and clinicians.⁷⁰ Several studies incorporated explainability features to help identify key parameters or input data that influence predictive output generation. It is important to note the technical limitations of existing explainability features such as saliency maps, however, as they do not render complex model behaviours fully transparent and may not be anchored in clinical knowledge.⁷¹ We recommend that future studies report the capabilities of explainability features more cautiously in order to establish realistic expectations for clinicians and AI developers that explainability features alone cannot make AI models fully interpretable or transparent, nor can they serve as an absolute guarantee for the trustworthiness of AI generated results. Effective trust-building amongst clinicians for better AI uptake may rely more on robust procedural safeguards rather than full mechanistic explainability, therefore, we encourage future studies to explore potential implementation strategies such as shadow deployment, recurring local validation, confidence-aware models and human-in-the-loop clinical pathways with clear AI accountability and governance structures.⁷²⁻⁷⁵

Only a minority of studies provided open-source access to their training data (5% of studies) and code (21% of studies), limiting independent external replication and validation. It may be difficult to

provide public access to identifiable medical datasets due to ethics and confidentiality constraints; however, we strongly recommend providing open access to AI model codes, trained weights and detailed cohort characteristics to facilitate external validation efforts. A related shortfall that limits model fairness and generalisability is the poor reporting of race and ethnicity statistics in included studies. Regrettably, only 40% of studies reported ethnicity and race characteristics. AI models exhibited race- and ethnicity-associated bias in glaucoma predictive performance,⁴⁵ therefore, future studies are encouraged to report detailed race and ethnicity statistics or infer ethnicity distribution from the underlying population where possible.

In light of the diverse AI approaches in the reviewed studies, which presented both innovative strengths and heterogeneity-associated challenges for clinical implementation, we recognise the importance of highlighting good AI practices for clinical tool development as well as advocating standardised, clinically focused reporting items. Research has been done towards building consensus around recommended reporting items for AI studies that perform clinical modelling, such as the MI-CLAIM checklist which outlines the minimum recommended reporting information for general clinical AI modelling.⁷⁶ Searches on the EQUATOR database returned no ophthalmology or glaucoma-specific reporting guidelines for AI studies, therefore, we drew on strengths and limitations noted in our qualitative synthesis and existing general clinical AI guidelines to propose a glaucoma-specific list of recommended practices and reporting items with a focus on study design, AI methodology and outcome reporting to aid future AI modelling studies (Table 4).

The findings from this study should be interpreted with a few limitations in mind. This review excluded conference abstracts due to risk of bias concerns and limited the search language to English, potentially missing the latest progress and research in other languages. We also did not explicitly assess the predictive performance of AI models in different subtypes of glaucoma, as many studies did not report subtype statistics to enable this fine-grained analysis. We believe the underlying pathophysiological differences between subtypes likely do not play a significant role in AI

performance, however, as AI model training for glaucoma prediction is more data-driven than mechanism-driven. Our search strategy focused primarily on glaucoma-related search terms. Future studies are encouraged to also include search terms for broader, multi-disease studies that may explore glaucoma progression prediction as a sub-section. We did not perform quantitative synthesis or head-to-head performance comparisons between studies due to marked between-study heterogeneity that may significantly confound results and limit the utility of quantitative analysis at this stage. We invite future studies to conduct more focused meta-analyses on this topic when sufficient studies with consistent study design and reporting become available. Health economics evaluation and policy-level planning for clinical implementation are beyond the scope of the current review but remain important considerations for successful clinical implementation. Cost-effectiveness of AI deployment has been explored for more mature use cases like glaucoma screening but remains poorly researched for progression prediction, therefore, we invite future studies to focus on these practical implementation issues.⁷⁷ Publication bias may also be a limitation to our study despite our efforts to cover multiple databases and preprint servers, which may render our analysis of AI performance over-optimistic.

In conclusion, current AI models demonstrated moderate to good performance mostly via internal validation across a range of progression prediction tasks including glaucoma conversion, biological progression and progression to surgery. Significant between-study heterogeneity and methodological shortfalls are prevalent, however, as current AI approaches frequently fail to contextualize AI outcomes with clinically relevant measures and exhibit inconsistent reporting of study design, cohort characteristics and AI outcomes at large. The overwhelming lack of external validation, explicit treatment effect modelling and limited AI transparency also remain key challenges to obtaining robust, generalisable AI performance and fostering trust towards AI amongst clinicians. In light of these challenges and the rapidly growing interest in AI applications for glaucoma, we proposed the first glaucoma-tailored recommended best practices and reporting items list which may also serve as guiding templates for AI studies in other clinical specialties. As clinicians currently have limited

options to tailor glaucoma management based on progression risk, AI tools may enable personalised glaucoma risk predictions and transform glaucoma care. We invite future studies to further refine AI study design and reporting and explore organisation and policy-level implementation strategies such as how AI models might integrate within existing eye care pathways to maximise safety, efficacy and cost-effectiveness.

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Methods

1. Protocol and registration

This review was conducted in accordance with the Preferred Reporting Items for Systematic Review and Meta-analysis of Diagnostic Test Accuracy Studies (PRISMA-DTA), which is an extension of the PRISMA guidelines designed for diagnostic test accuracy (DTA) reviews.⁷⁸ The review protocol was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO ID: CRD42024524655).

2. Eligibility criteria

The primary objective of this review was to examine the predictive accuracy of AI models for glaucoma progression and explore heterogeneity within current AI approaches. Multiple definitions of AI and glaucoma progression are presently used in research and clinical contexts. To maximally capture relevant studies without introducing excessive heterogeneity for data synthesis, we elected to use broad definitions for key components of the research question whilst maintaining some specificity to ensure clinical relevance. The PICO (population, index test, comparator/reference standard and outcome) table outlined these definitions for the primary objective (Table 1). Notably, all clinically relevant definitions of glaucoma progression were considered for the reference standard, including functional and structural measures as well as clinical records of critical progression events. This reflects the concurrent use of diverse measures of glaucoma disease activity in routine care.

Additionally, we have specified predicting glaucoma progression as the forecasting of future glaucoma disease activity using current or past data which excluded studies on the detection of current or past progression. Diverse quantitative outcome measures were considered for this review in light of the inclusion of multiple reference standards, including binary prediction metrics, numeric prediction errors and survival accuracy indices. The inclusion criteria were therefore primary studies since 2014 with English full text which report quantitative outcomes of AI models that predict

glaucoma progression in glaucoma and/or suspect patients, using the relevant definitions outlined in Table 1. The time limit of 2014 was guided by pilot searches and covered most relevant research with AI models. Review articles were excluded as well as conference abstracts, which lack key study details necessary for risk of bias assessment and data synthesis. Preprints were included with caution as grey literature. They were subject to the same screening and risk of bias assessment procedures as peer-reviewed journals whilst maintaining a preprint label, and any apparent discrepancy from the peer-reviewed group was used to determine potential exclusion from the review and subsequent meta-analysis.

3. Search strategy

A comprehensive search strategy was developed using Medical Subject Headings (MeSH) terms and keywords, including variations of “artificial intelligence”, “glaucoma”, “progression” and “prediction”. Searches were conducted across MEDLINE, Embase (including preprints from MedRxiv and bioRxiv), Web of Science Core Collection, Cochrane CENTRAL as well as arXiv to cover eligible grey literature. The full search queries are included in Supplementary section 2.

4. Study selection

Results from literature searches were imported into Covidence, where two independent reviewers (YL and LF) screened the titles and abstracts of unique entries in a masked fashion against the inclusion and exclusion criteria after removing duplicates. Disagreements were resolved with a third, expert reviewer (AW or GL). The two independent, masked reviewers (YL and LF) then reviewed the full text of selected entries, and eligible studies underwent data extraction in a masked fashion as well. The reference lists of eligible studies were also screened for missing studies, and a follow-up search was conducted in January 2025 to capture recent publications.

5. Data extraction

Data extraction was performed in Covidence with a pre-defined extraction form, which included publication details, cohort details (size, source, location, eligibility criteria, characteristics, ethnicity reporting), AI model details (input modality, training settings, model design), primary outcome (quantitative performance metrics, contingency tables) and secondary outcomes (external validation, open-source status, explainability features). For studies that reported multiple models, all eligible models were extracted as correlated entries to minimise selection bias. For studies that trained the same model designs with different combinations of input data, we selected the most optimised input combination associated with the highest test performance which aligns with the AI model development process, where different input combinations primarily serve as exploratory steps in model optimisation and the optimal model is reported. Disagreements in the extracted data were resolved via discussion and consensus amongst all reviewers.

6. Risk of bias assessment

The risk of bias and quality of each study was assessed by two reviewers (YL and LF) independently during data extraction using the Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) tool. Risk of bias outcomes were visualised using the *robvis* package in R (version 0.3.0.900).⁷⁹ Publication merging was conducted prior to assessing risk of bias, where publications with identical cohort characteristics and consistent data sources were merged as a single study to highlight potential correlations between their outcomes. Risk of bias was then assessed and reported on the study level, where any inconsistency between merged publications received an “unclear” or “high risk” assessment depending on the risk of bias of the underlying publications. Publication bias was qualitatively assessed by evaluating the coverage of our search strategy and analysing any notable disparities in performance outcomes based on cohort sizes.

Data availability

All data collected and generated from this systematic review, including raw data not detailed in the supplementary materials, may be requested from the corresponding author upon reasonable request.

Code availability

Not applicable

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Author contributions

Y.G.L. and L.F. performed the literature searches, screened the titles, abstracts and full-text records of search results, and extracted relevant data from eligible studies. G.L. and A.J.R.W. adjudicated on the differences arising from the literature screening and data extraction stages. Y.G.L. performed data analysis and drafted the manuscript. A.T.P. provided guidance on the statistical methods and data analysis. A.T.P., G.L. and A.J.R.W. reviewed and provided general feedback to the manuscript.

Competing interests

The authors declare no competing interests.

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Figures and Tables

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) Flow

Diagram for Study Selection

This diagram details the literature search and selection process for this study. 4108 records were identified from initial database searches, and 30 records were identified from the Cochrane CENTRAL clinical trial register. 2767 records progressed to title and abstract screening after removing duplicates, and 129 were eligible for full text screening. 46 reports of 43 studies were deemed eligible for review.

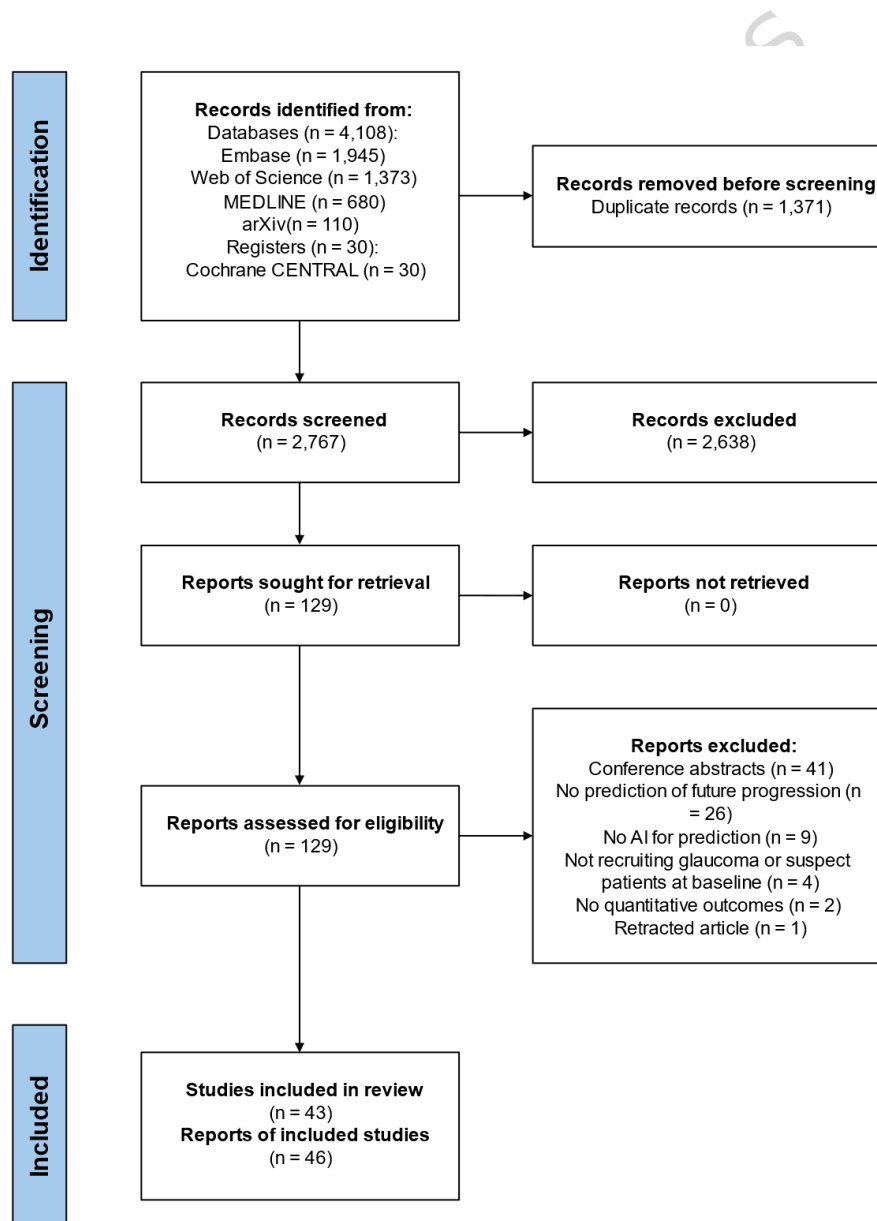


Figure 2. Summary scatter plot of sensitivity and specificity metrics for binary classification models that predict glaucoma conversion or progression

A. Summary plot of binary classification performance of AI models that predict conversion from OHT/suspect to glaucoma. Each data point represents a single model with an ID linking it to its original study. Data point colours represent different reference standards for progression, and data point shape represents input modality of AI models. Data point size represents size of the testing dataset, which is a surrogate marker for overall dataset size.

B. Summary plot of binary classification performance of AI models that predict progression in diagnosed glaucoma patients using biologically grounded reference standards. Each data point represents a single model with an ID linking it to its original study. Data point colours represent different reference standards for progression, and data point shape represents input modality of AI models. Data point size represents size of the testing dataset, which is a surrogate marker for overall dataset size.

C. Summary plot of binary classification performance of AI models that predict progression in diagnosed glaucoma patients using clinical event-based reference standards. Each data point represents a single model with an ID linking it to its original study. Data point colours represent different reference standards for progression, and data point shape represents input modality of AI models. Data point size represents size of the testing dataset, which is a surrogate marker for overall dataset sizes.

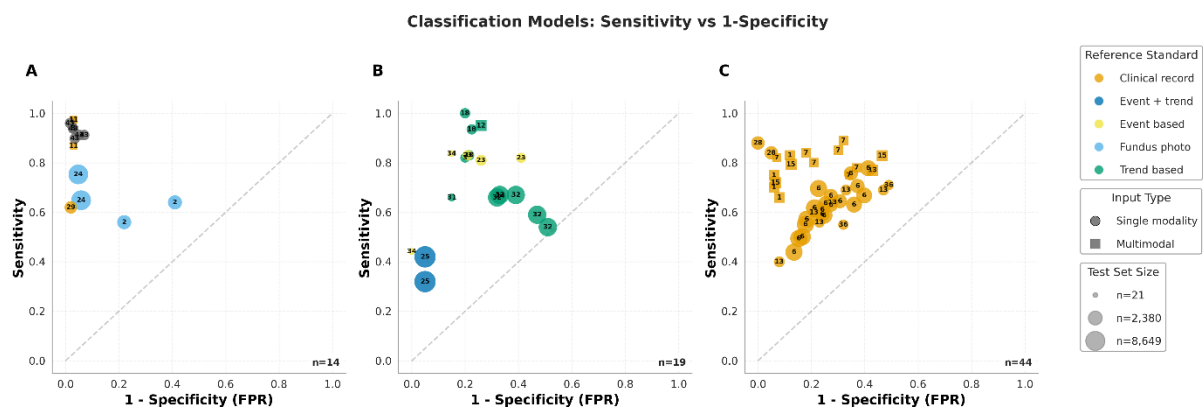


Figure 3. Summary scatter plot of mean absolute errors (MAE) and root mean square errors (RMSE) for numeric regression models that predict functional or structural glaucoma parameters

A. Summary plot of the predictive errors (MAE and RMSE) for numeric regression models that predict pointwise deviations on 24-2 and 30-2 visual fields. Each data point represents a single model from a study on the y-axis. Data point colours represent the glaucoma parameter being predicted, and data point shape represents the type of predictive error being reported (MAE or RMSE). The data point sizes are relative to the size of the test dataset which is a surrogate marker for overall dataset sizes.

B. Summary plot of the predictive errors (RMSE) for numeric regression models that predict future intraocular pressures (IOP). Each data point represents a single model from a study on the y-axis. Data point colours represent the glaucoma parameter being predicted, and data point shape represents the type of predictive error being reported (MAE or RMSE). The data point sizes are relative to the size of the test dataset which is a surrogate marker for overall dataset sizes.

C. Summary plot of the predictive errors (MAE and RMSE) for numeric regression models that predict future mean deviations (MD) of visual fields. Each data point represents a single model from a study on the y-axis. Data point colours represent the glaucoma parameter being predicted, and data point shape represents the type of predictive error being reported (MAE or RMSE). The data point sizes are relative to the size of the test dataset which is a surrogate marker for overall dataset sizes.

D. Summary plot of the predictive errors (MAE and RMSE) for numeric regression models that predict future pattern standard deviations (OSD) of visual fields. Each data point represents a single model from a study on the y-axis. Data point colours represent the glaucoma parameter being predicted, and data point shape represents the type of predictive error being reported (MAE or RMSE). The data point sizes are relative to the size of the test dataset which is a surrogate marker for overall dataset sizes.

E. Summary plot of the predictive errors (MAE and RMSE) for numeric regression models that predict future visual field indices (VFI) on visual fields. Each data point represents a single model from a study on the y-axis. Data point colours represent the glaucoma parameter being predicted, and data point shape represents the type of predictive error being reported (MAE or RMSE). The data point sizes are relative to the size of the test dataset which is a surrogate marker for overall dataset sizes.

F. Summary plot of the predictive errors (MAE and RMSE) for numeric regression models that predict future retinal nerve fibre layer (RNFL) thickness or thinning rates. Each data point represents a single model from a study on the y-axis. Data point colours represent the glaucoma parameter being predicted, and data point shape represents the type of predictive error being reported (MAE or RMSE). The data point sizes are relative to the size of the test dataset which is a surrogate marker for overall dataset sizes.

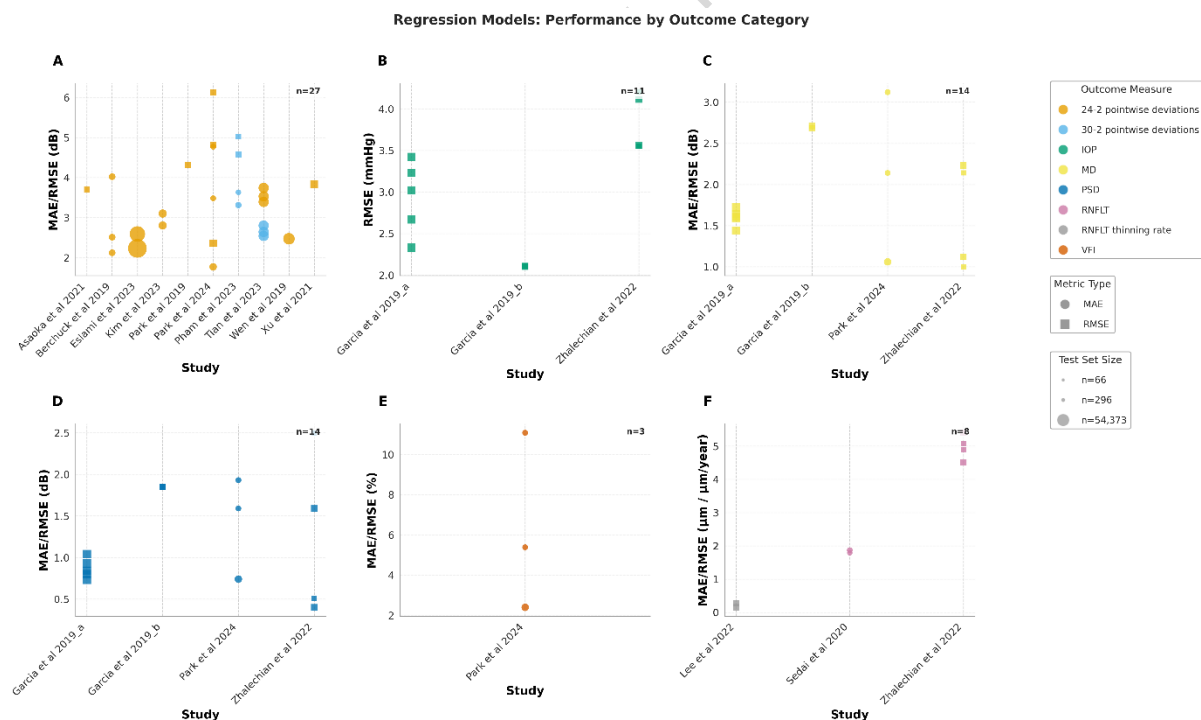


Figure 4. Risk of bias and applicability assessment of included studies

The risk of bias and applicability of each study was assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) tool. An unclear rating was given when insufficient information was reported to make a judgement towards either low or high risk. Figure 4A and 4B report outcomes for the four risk of bias domains. Figures 4C and 4D report outcomes for the three applicability domains.

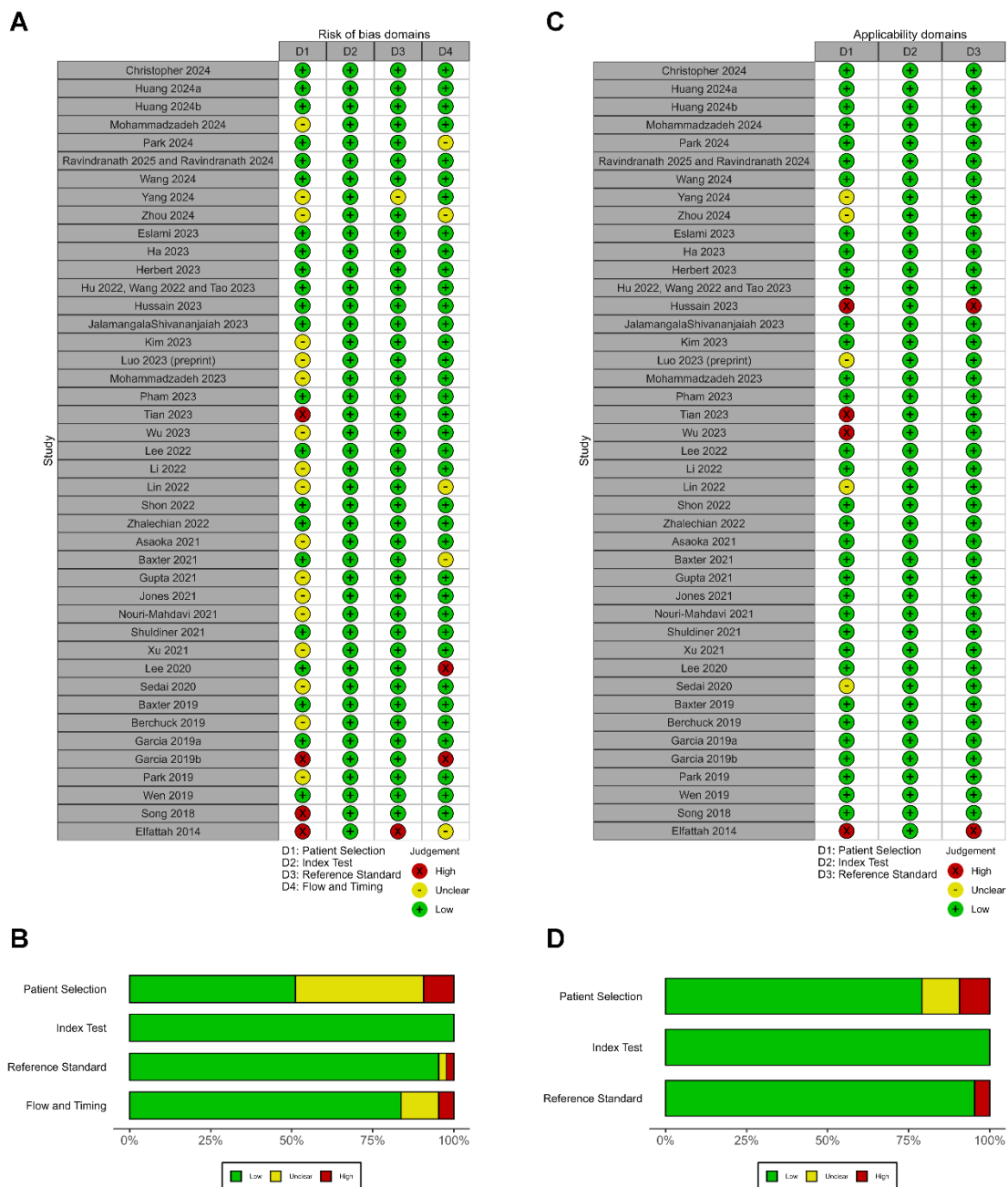


Table 1. PICO (population, index test, comparator/reference standard and outcome) table for the primary objective

ML: machine learning; DL: deep learning; VF: visual field; OCT: optical coherence tomography; AUC: area under the curve; RMSE: root mean square error; MAE: mean absolute error; C-index: concordance index.

PICO Component	Definition (Inclusion Criteria)	Exclusion criteria
Population	Patients with any type of glaucoma and glaucoma suspects	Healthy individuals Other eye conditions
Index Test	AI models (ML and DL models) that predict future glaucoma progression using current or past data	Traditional statistical regression methods AI models that detect current or past progression
Comparator (Reference Standard)	All clinically relevant definitions of glaucoma progression: - Functional measures such as VF metrics - Structural measures such as OCT parameters - Clinical records of critical events such as surgical procedures	Clinically irrelevant definitions of progression Ocular parameters unrelated to progression
Outcome	All quantitative metrics: - Binary prediction metrics such as AUC, sensitivity and specificity, - Numeric prediction errors such as RMSE and MAE - Survival accuracy indices such as C-index	Qualitative metrics Metrics unrelated to progression

Table 2. Study details, design and cohort characteristics of included studies

OHT: ocular hypertension. UWHVF: University of Washington Humphrey Visual Fields. CNN: convolutional neural network. RNN: recurrent neural network.

NTG: normal tension glaucoma. HTG: high tension glaucoma. POAG: primary open angle glaucoma.

Study	ID	Study Location	Design	Cohort Type	Cohort Size	Glaucoma Severity (baseline or overall)	Mean Age (Years $\pm$ SD)	Female Male Ratio	Race Ethnicity Reporting
Christopher 2024	1	US	Cohort study	Glaucoma	1179	Overall MD (95% CI): -7.17 (-7.70 - -6.63) - Training, Surgery -1.40 (-1.85 - -0.94) - Training, No Surgery -8.39 (-9.74 - -7.05) - Testing, Surgery -1.19 (-2.18 - -0.20) - Testing, No Surgery	68.4 $\pm$ 9.92 (Training, surgery) 63.2 $\pm$ 10.79 (Training, no surgery) 66.3 $\pm$ 10.45 (Testing, surgery) 59.8 $\pm$ 10.35 (Testing, no surgery)	1.07 (Training, surgery) 1.6 (Training, no surgery) 2.0 (Testing, surgery) 1.3 (Testing, no surgery)	Yes
Huang 2024_a	2	US	Cohort study	OHT/Suspect	1504	Overall IOP (SD): 18.5 (3.9) - Unconverted 20.0 (5.8) - Converted Overall MD (SD): -0.3 (2.4) - Unconverted -1.0 (2.1) - Converted Overall PSD (SD): 2.1 (1.2) - Unconverted 2.4 (1.3) - Converted	65.2 $\pm$ 9.6 (Unconverted) 66.7 $\pm$ 9.5 (Converted)	1.5 (Unconverted) 0.75 (Converted)	Yes
Huang 2024_b	3	US	Cohort study	Glaucoma	784	Baseline MD (SD): -2.52 (4.54) - No progression -2.64 (3.99) - Moderate progression -4.06 (4.12) - Rapid progression	66.5 $\pm$ 12.7 (No progression) 66.6 $\pm$ 11.5 (Moderate progression) 67.7 $\pm$ 12.3 (Rapid progression)	1.5 (Total)	No
Mohammadza deh 2024	4	US	Cohort study	Glaucoma	1764	Baseline MD (SD): -3.2 (4.9) - Stable	59.7 $\pm$ 12.8 (Stable) 61.6 $\pm$ 10.6 (Progressing)	1.34 (Stable) 1.29 (Progressing)	Yes

Study	ID	Study Location	Design	Cohort Type	Cohort Size	Glaucoma Severity (baseline or overall)	Mean Age (Years $\pm$ SD)	Female Male Ratio	Race Ethnicity Reporting
						-4.0 (5.0) – Progressing			
Park 2024	5	South Korea	Cohort study	All	13348	Overall MD (SD): -7.81 (8.74) – Training -5.91 (7.47) – Test	56.9 $\pm$ 17.0 (Training) 51.4 $\pm$ 19.0 (Test)	NA	No
Ravindranath 2025 and Ravindranath 2024	6	US	Cohort study	Glaucoma	39090	Max IOP, either eye (SD): 19.3 (7.6) – Total 18.7 (7.1) – No Surgery 22.3 (9.1) – Surgery	70.1 $\pm$ 14.6 (Total) 70.8 $\pm$ 14.4 (No surgery) 66.7 $\pm$ 15.3 (Surgery)	1.20 (Total) 1.21 (No surgery) 1.16 (Surgery)	Yes
Wang 2024	7	US	Cohort study	Glaucoma	4038	Overall Median MD (IQR): -4.9 (-12.8, -2) [0-0.25] years -3.9 (-8.5, -1.5) [0.25-0.5] years -3.5 (-8.2, -1.3) [0.5-1] years -3.5 (-7.1, -1.2) [1-2] years -3.1 (-6.8, -1.0) [2-3] years -3.0 (-6.4, -0.8) [3-4] years -2.7 (-6.0, -0.8) [4-5] years	NA	1.22 [0-0.25] years 1.30 [0.25-0.5] years 1.31 [0.5-1] years 1.39 [1-2] years 1.33 [2-3] years 1.43 [3-4] years 1.28 [4-5] years	No
Yang 2024	8	China	Cohort study	OHT/Suspect	NA (405 eyes)	NA	NA	NA	No
Zhou 2024	9	US	Cohort study	OHT/Suspect	1436	NA	NA	NA	No
Eslami 2023	10	US	Cohort study	All	21120	Overall MD (SD): -3.95 (5.66) – Total Baseline MD (SD): -2.74 (4.54) – MWen Cohort -1.4 (3.49) – MPark Cohort	62.94 $\pm$ 15.07 (Total) 64.06 $\pm$ 13.50 (CNN/MWen Cohort) 62.64 $\pm$ 11.53 (RNN/MPark Cohort)	1.28 (Total)	No

Study	ID	Study Location	Design	Cohort Type	Cohort Size	Glaucoma Severity (baseline or overall)	Mean Age (Years $\pm$ SD)	Female Male Ratio	Race Ethnicity Reporting
Ha 2023	11	South Korea	Cohort study	OHT/Suspect	210	Baseline IOP (SD): 14.8 (2.9) – Total 16.7 (1.9) - Training, conversion to NTG 12.9 (2.5) - Training, no conversion 16.9 (1.8) - Testing, conversion to NTG 12.6 (2.0) Testing, no conversion	55.8 $\pm$ 9.5 (Total) 50.0 $\pm$ 8.1 (Training, conversion to NTG) 61.6 $\pm$ 6.8 (Training, no conversion) 50.6 $\pm$ 7.6 (Testing, conversion to NTG) 61.6 $\pm$ 7.8 (Testing, no conversion)	0.74 (Total) 0.75 (Training, conversion to NTG) 0.79 (Training, no conversion) 0.75 (Testing, conversion to NTG) 0.59 (Testing, no conversion)	No
Herbert 2023	12	US	Cohort study	All	2962	Overall MD (SD) -3.42 (4.29) - Total -3.15 (4.04) - Non-rapid -7.83 (5.61) - Rapid progressors	65.87 $\pm$ 12.40 (Total) 65.66 $\pm$ 12.40 (Non-rapid) 69.30 $\pm$ 11.94 (Rapid progressors)	1.29 (Total) 1.31 (Non-rapid) 1.01 (Rapid progressors)	No
Hu 2022, Wang 2022 and Tao 2023	13	US	Cohort study	Glaucoma	4512	Overall IOP (SD) 18.3/18.8 (12.3/19.1) – Right/Left Eye, Total 20.1/21.8 (27.7/45.8) – Right/Left Eye, Surgery 18.0/18.3 (6.2/6.5) – Right/Left Eye, No surgery	65.0 $\pm$ 17.9 (Total) 65.0 $\pm$ 18.1 (Surgery) 64.8 $\pm$ 17.0 (No surgery)	1.01 (Total) 1.04 (Surgery) 0.88 (No surgery)	Yes
Hussain 2023	14	Lithuania	Cohort study	Glaucoma	86	Baseline MD (SD): -14.68 (8.43)	67.28 $\pm$ 8.92	0.89	No
JalamangalaS hivananjaiah 2023	15	US	Cohort study	Glaucoma	3469	Overall IOP (SD): 15.4/15.4 (6.1/6.2) Right/Left Eye, Total 15.1/15.3 (5.0/5.7) Right/Left Eye, No surgery 15.5/15.4 (6.8/6.6) Right/Left Eye, Surgery	67 $\pm$ 18 (Total) 69 $\pm$ 17 (No surgery) 66 $\pm$ 18 (Surgery)	1.04 (Total) 1.08 (No surgery) 0.93 (Surgery)	Yes

Study	ID	Study Location	Design	Cohort Type	Cohort Size	Glaucoma Severity (baseline or overall)	Mean Age (Years $\pm$ SD)	Female Male Ratio	Race Ethnicity Reporting
Kim 2023	16	South Korea	Cohort study	All	4593	Baseline MD (SD) -5.83 (6.21) - Total -5.77 (6.16) - Training -6.19 (6.44) - Testing	53.96 $\pm$ 16.04 (Total) 54.11 $\pm$ 15.88 (Training) 53.13 $\pm$ 16.85 (Testing)	NA	No
Luo 2023 (preprint)	17	US	Cohort study	Glaucoma	500	NA	NA	0.79	Yes
Mohammadzadeh 2023	18	US	Cohort study	Glaucoma	2259	Baseline MD (SD) -3.3 (4.9) - Stable -3.6 (4.8) - Progressing	59.5 $\pm$ 12.9 (Stable) 61.0 $\pm$ 11.4 (Progressing)	1.34 (Stable) 1.52 (Progressing)	Yes
Pham 2023	19	South Korea	Cohort study	Glaucoma	316	Overall MD (SD) -6.4 (5.1) - Training -2.4 (6.0) - Testing	57.4 $\pm$ 12.8 (Training) 50.9 $\pm$ 12.4 (Testing)	NA	No
Tian 2023	20	US	Cohort study	All	9038	NA	NA	1.4 (62K+ dataset) 1.2 (UWHVF dataset)	No
Wu 2023	21	US	Cohort study	Glaucoma	NA (1706 eyes)	NA	NA	NA	No
Lee 2022	22	South Korea	Cohort study	Glaucoma	712	Baseline MD (SD): -5.00 (6.11)	54.1 $\pm$ 14.5	1.02	No
Li 2022	23	China	Cohort study	OHT/Suspect	2219	Overall MD (SD): -2.6 (4.1) - Training -2.6 (4.8) - Validation -2.2 (3.6) - External Test 1 -2.7 (4.5) - External Test 2	44.7 $\pm$ 14.3 (Training) 42.8 $\pm$ 13.2 (Validation) 41.7 $\pm$ 15.2 (External Test 1) 45.1 $\pm$ 13.9 (External Test 2)	1.44 (Training) 1.86 (Validation) 1.26 (External Test 1) 1.32 (External Test 2)	No
Lin 2022	24	US	Cohort study	OHT/Suspect	3273	NA	NA	NA	No
Shon 2022	25	South Korea	Cohort study	Glaucoma	6047	Baseline MD (SD): -4.67 (5.92) - Stable -9.09 (6.07) - Progressing	56.6 $\pm$ 11.7 (Stable) 57.8 $\pm$ 12.8 (Progressing)	NA	No
Zhalechian 2022	26	US	Cohort study	Glaucoma	362	Baseline MD (SD): -0.5 (1.5) - Suspects -4.7 (5.9) - POAG	69.7 $\pm$ 11.0 (Suspects) 71.7 $\pm$ 10.2 (POAG)	1.64 (Suspects) 1.10 (POAG)	Yes

Study	ID	Study Location	Design	Cohort Type	Cohort Size	Glaucoma Severity (baseline or overall)	Mean Age (Years $\pm$ SD)	Female Male Ratio	Race Ethnicity Reporting
Asaoka 2021	27	Japan	Cohort study	Glaucoma	676	Overall MD (SD): -6.2 (7.1) - Training -4.9 (4.6) - Testing	60.7 $\pm$ 13.5 (Training) 61.2 $\pm$ 10.4 (Testing)	1.00 (Training) 1.21 (Testing)	No
Baxter 2021	28	US	Cohort study	Glaucoma	1231	NA	69.1 $\pm$ 10.5	1.34	Yes
Gupta 2021	29	US	Cohort study	OHT/Suspect	438	Baseline IOP (SD): 24.9 (2.4) - Unconverted 25.4 (2.6) - Converted	54.4 $\pm$ 7.3 (Unconverted) 58.8 $\pm$ 9.8 (Converted)	1.56 (Unconverted) 0.57 (Converted)	Yes
Jones 2021	30	US	Cohort study	Glaucoma	295	Baseline MD (SD): -7.4 (5.5) - Total -8.4 (5.4) - Black cohort -6.5 (5.3) - White cohort -4.4 (5.7) - Asian cohort	NA	1.09 (Total) 1.61 (Black cohort) 0.69 (White cohort) 2.00 (Asian cohort)	Yes
Nouri-Mahdavi 2021	31	US	Cohort study	Glaucoma	104	Baseline MD (SD): -8.4 (5.5) - Total -8.8 (6.2) - Stable -7.5 (5.6) - Progressing	67.3 $\pm$ 8.2 (Total) 66.6 $\pm$ 8.0 (Stable) 69.1 $\pm$ 7.7 (Progressing)	1.74 (Total) 1.53 (Stable) 2.29 (Progressing)	Yes
Shuldiner 2021	32	US	Cohort study	Glaucoma	14217	Baseline MD (SD): -5.2 (6.0) - Total -5.0 (6.0) - Non-rapid -7.1 (5.7) - Rapid progressors	62.4 $\pm$ 13.3 (Total) 61.9 $\pm$ 13.6 (Non-rapid) 68.6 $\pm$ 12.8 (Rapid progressors)	NA	No
Xu 2021	33	Japan	Cohort study	Glaucoma	592	NA	NA	NA	No
Lee 2020	34	South Korea	Cohort study	Glaucoma	155	Baseline IOP (SD): 14.3 (3.4) - Training, no progression 14.1 (2.5) - Training, progression 13.4 (2.8) - Testing, no progression 14.4 (2.4) - Testing, progression	34.49 $\pm$ 4.18 (Training, no progression) 34.65 $\pm$ 4.64 (Training, progression) 33.17 $\pm$ 4.34 (Testing, no progression) 31.52 $\pm$ 7.69 (Testing, progression)	0.72 (Training, no progression) 0.87 (Training, progression) 0.60 (Testing, no progression) 0.62 (Testing, progression)	No

Study	ID	Study Location	Design	Cohort Type	Cohort Size	Glaucoma Severity (baseline or overall)	Mean Age (Years $\pm$ SD)	Female Male Ratio	Race Ethnicity Reporting
Sedai 2020	35	US	Cohort study	All	1089	Baseline MD (SD): -1.72 (3.22) - Suspects -5.49 (7.05) - Glaucoma	59.99 $\pm$ 13.02 (Suspects) 63.29 $\pm$ 12.64 (Glaucoma)	1.29 (Suspects) 1.16 (Glaucoma)	No
Baxter 2019	36	US	Cohort study	Glaucoma	385	NA	73.24 $\pm$ 11.88 (No Surgery) 73.09 $\pm$ 12.60 (Surgery)	1.059 (Total) 1.153 (No Surgery) 0.955 (Surgery)	Yes
Berchuck 2019	37	US	Cohort study	All	3832	Baseline MD (SD): -3.55 (5.71) - Total -3.57 (5.72) - Training -3.76 (6.12) - Validation -3.23 (5.17) - Test	60.94 $\pm$ 15.66 (Total) 61.04 $\pm$ 15.77 (Training) 60.38 $\pm$ 15.9 (Validation) 60.67 $\pm$ 14.47 (Test)	NA	No
Garcia 2019_a	38	US	Cohort study	OHT/Suspect	1407	Baseline MD (SD): 0.0 (1.4)	57.5 $\pm$ 9.6	1.35	Yes
Garcia 2019_b	39	Japan	Cohort study	Glaucoma	864	Baseline MD (SD): -6.2 (5.9) - NTG -7.5 (5.6) - HTG	63.4 $\pm$ 10.5 (NTG) 66.1 $\pm$ 10.9 (HTG)	0.65 (NTG) 1.07 (HTG)	Yes
Park 2019	40	South Korea	Cohort study	All	1122	Baseline MD (SD): -7.02 (6.09) - Training -6.35 (5.20) - Testing	58.9 $\pm$ 16.2 (Training) 63.2 $\pm$ 14.4 (Testing)	0.88 (Testing)	No
Wen 2019	41	US	Cohort study	All	4875	Baseline MD (SD): 61.9 (14.8) - Total 62.0 (14.8) - Training 61.8 (15.1) - Testing	61.9 $\pm$ 14.8 (Total) 62.0 $\pm$ 14.8 (Training) 61.8 $\pm$ 15.1 (Testing)	1.18 (Total) 1.16 (Training) 1.26 (Testing)	No
Song 2018	42	US	Cohort study	Glaucoma	134	Baseline MD (SD): -3.50 (6.04)	61.9 $\pm$ 1.4	1.045	No
Elfattah 2014	43	Egypt	Cohort study	OHT/Suspect	398	Overall IOP (SD): 26.67 (0.149)	55.93 $\pm$ 10.071	NA	No

Table 3 Summary Table of Quantitative Performance Outcome Ranges Stratified by Predictive Task and Reported Outcome

AUC: area under the curve. IOP: intraocular pressure. MD: mean deviation. PSD: pattern standard deviation. VFI: visual field index. RNFLT: retinal nerve fibre thickness. VF: visual field. RMSE: root mean square error. MAE: mean absolute error.

Predictive Task	Reported Outcome	Number of Studies	Range
Binary prediction of glaucoma conversion	AUC	6	0.620-0.998
	Sensitivity	6	0.560-0.974
	Specificity	6	0.590-0.983
Binary prediction of glaucoma progression	AUC	19	0.650-0.990
	Sensitivity	19	0.320-1.000
	Specificity	19	0.490-1.000
Prediction of numeric variables	IOP	3	RMSE: 2.10-4.20 mmHg
	MD	4	RMSE: 1.99-2.71 dB MAE: 1.06-3.12 dB
	PSD	4	RMSE: 0.40-2.50 dB MAE: 0.74-1.93 dB
	VFI	1	MAE: 2.39%-11.08%
	RNFLT	2	RMSE: 4.51-5.40 μ m MAE: 1.79-1.87 μ m
	RNFLT thinning rate	1	RMSE: 0.152-0.276 μ m/year
	24-2 Pointwise Deviations	9	RMSE: 2.35-6.13 dB MAE: 1.77-4.77 dB
	30-2 Pointwise Deviations	2	RMSE: 4.58-5.02 dB MAE: 2.54-3.63 dB
Prediction of time to progression event	Concordance Index	2	0.543-0.775

Table 4. Recommended practices and reporting items for the development and evaluation of AI models predicting glaucoma progression

POAG: primary open angle glaucoma. NTG: normal tension glaucoma. MD: mean deviation. IOP: intraocular pressure. VF: visual field. OCT: optical coherence tomography. SD: standard deviation. IQR: interquartile range. CNN: convolutional neural network. ViT: vision transformer. GBM: gradient boosting machine. SHAP: SHapley Additive exPlanations. LIME: Local Interpretable Model-Agnostic Explanations.

Domain	Reporting Items	Recommended Practice
Study Design	Study Type	- Specify if the study is retrospective or prospective. - Prospective and randomised trial designs should follow existing AI trial guidelines such as SPIRIT-AI or CONSORT-AI
	Research Question	Clearly define the research question, including the target population, ground truth definition, AI methodology and the clinical task at minimum.
	Target Population	- Specify the target population of the AI models, which should at minimum include disease subtype (e.g. POAG, NTG), age range (e.g. adults versus over 40s) and disease severity (e.g. early versus advanced). - Reporting of more fine-grained qualifiers such as geographical location, ethnicity context, treatment status and concomitant ocular conditions is strongly recommended. - Justification of the target population should be included and based on study cohort characteristics, as discussed below.
	Ground Truth Definition	- The definition for the ground truth (e.g. conversion to glaucoma or glaucoma progression) should be clearly defined based on clinically relevant and ideally quantitative or objective criteria. - For longitudinal modelling, the baseline must be clearly and consistently defined. - For trend-based analyses, a minimum number of observations should be defined to ensure statistically robust trend calculation. - For event-based analyses, the event and time to event should be consistently and objectively defined. - If multiple definitions are concurrently in use clinically, we recommend exploring multiple clinically relevant definitions as separate experiments.
	Predictive Task	- The predictive task trained and evaluated on the AI models should be clearly defined with reference to the appropriate ground truth definitions. - For cross-sectional or classification tasks, we recommend opting for more fine-grained categories beyond binary classification where possible. - For longitudinal and regression tasks, we recommend the use of standardised prediction timeframes in yearly and half-yearly steps.

Domain	Reporting Items	Recommended Practice
Study Cohort	Data Source	1. The data source for AI training and validation should be clearly reported. 2. Key information about the data source including country, centre and type (primary/tertiary/research) should be reported at minimum. 3. Multi-region and multicentre sources covering both primary and tertiary institutions are ideal.
	Recruitment Method	1. Recruitment or data collection method should be clearly stated. 2. Consecutive or randomised recruitment is preferred over convenient sampling.
	Eligibility Criteria	1. The inclusion and exclusion criteria should be specified, ideally with objective, quantitative standards. 2. Eligibility criteria should reflect typical patient distribution within the underlying data source and the intended target population. 3. Any inclusions and exclusions that may significantly alter the data distribution beyond the study-specific clinical norms should be justified. 4. If eligibility criteria include subjective human grading, at least two independent reviewers should conduct masked screening in parallel with robust difference adjudication processes.
	Cohort Characteristics	1. Detailed cohort characteristics should be reported for the overall study data as well as significant subgroups, such as when stratified by disease staging or AI training and testing. 2. The distribution of baseline age, disease/label prevalence, disease severity (e.g. MD or IOP), follow-up period or longitudinal series length should be reported at minimum. 3. The distribution of the specific data modality used for AI training and evaluation (e.g. VF or OCT) should also be reported. 4. For trend-based analyses, the distribution of target trends (e.g. MD slope) should be reported. 5. For event-based analyses, the event rate and distribution of time to event should be reported. 6. When reporting distributional statistics, only report mean and SD for normally distributed data, otherwise consider reporting median and IQR instead. 7. Reporting of gender ratios are highly recommended where possible. 8. Reporting of detailed ethnicity and race statistics is highly recommended where possible. 9. Reporting of baseline treatment statistics is also encouraged along with statistics about prevalence of concomitant ocular condition (e.g. cataract, high myopia) that may be relevant for the AI task.
AI Training	AI Model Design	1. Clearly report the design of AI models including the overall architecture (e.g. CNN, ViT, GBM), input layers, input modality and output modality. 2. We recommend including simpler, more interpretable models as baseline performance comparisons when proposing new, advanced AI models to justify the need for the additional complexity in model design. 3. Models capable of leveraging multimodal inputs are recommended over single modality models, however, single modality models should still be reported as comparative baselines. 4. We highly recommend incorporating modules and techniques that account for latent and heterogeneous treatment effects as part of the AI model design.

Domain	Reporting Items	Recommended Practice
		5. We also recommend the adoption of causal AI models for clinical outcome prediction in addition to correlational models.
	Data Pre-processing	1. Specify any pre-processing of raw data before AI training and evaluation, which could include both data augmentation and quality control steps.
	Train/Validation/Test Split	1. Specify the method of generating the training, validation and test datasets. 2. Patient-level division is recommended to minimise the risk of data leakage between the training and evaluation datasets. 3. Report the prevalence of positive and negative labels or events and specify strategies used to address class imbalance.
	Hyperparameter Configuration	1. Specify the training hyperparameters including loss functions, optimizer, initial learning rate and schedules, epochs and any early stopping criteria, batch sizes and regularization. 2. We recommend setting a random seed during training to facilitate future external replication. 3. Reporting of the training and validation accuracy and loss curves is strongly recommended to demonstrate effective model learning with minimal overfitting.
	Training Environment	1. Specify the software and hardware environment for the AI training and evaluation process and include the typical training and inference time.
	Missing data	1. Specify how missing data is handled during AI training and testing. 2. We recommend testing different handling strategies to probe the effect of missing data on AI performance and select the optimal strategy for the target clinical task.
AI Evaluation	Evaluation Strategy	1. Clearly report the evaluation method. Cross-validation is recommended over a single validation dataset. 2. External validation is strongly recommended which may use data collected from external sites and/or from different equipment.
	Evaluation Outcomes	1. We recommend the reporting of detailed evaluation metrics based on the predictive task. For classification tasks, reporting of the AUC (and the optimal threshold), accuracy, sensitivity, specificity and confusion matrices is recommended at minimum. 2. For regression tasks, we recommend the reporting of predictive errors (e.g. MAE, RMSE) as well as their distributional statistics (SD, IQR). 3. For survival analyses, we recommend the reporting of concordance indices as well as hazard ratios and odds ratios between risk groups. 4. For all quantitative performance metrics, distributional statistics or confidence intervals should be reported where possible. 5. We recommend performing separate AI evaluations stratified by disease severity, prediction time horizon where possible. 6. We recommend reporting AI performance outcomes stratified by gender and ethnicity where possible.

Domain	Reporting Items	Recommended Practice
	Clinical Context	1. We strongly recommend the reporting of clinically relevant performance outcomes with the target patient cohort and disease prevalence in mind. 2. For classification tasks, we recommend the reporting of PPV and NPV with reference to the underlying disease or label prevalence in the target population. 3. For regression tasks, we recommend the reporting of prediction confidence intervals and recommended decision thresholds (e.g. $P < 0.05$). 4. For survival tasks, we recommend the reporting of risk stratified median survival times or event rates in relation to the baseline survival time or event rate in the target population. 5. We also recommend the calculation and reporting of NNT for all predictive task types.
	Explainability	1. We recommend the use of explainability models and explainability features (e.g. SHAP, feature important, saliency maps, LIME) to evaluate model behaviour. 2. Specify the explainability method used and report the purpose and capability of the explainability method (e.g. technical debug, feature selection). 3. We recommend against making inflated claims about mechanistic interpretability of black box AI models through explainability visualizations.
	External Reproducibility	1. We strongly recommend making the AI model weights and training codes open access via secure research code and data repositories such as GitHub and Zenodo. 2. We recommend the release of de-identified datasets used in model training and evaluation. If this is not feasible due to ethics or confidentiality constraints, we also recommend the release of a small scale, de-identified or synthetic validation dataset that could enable replication by external researchers and be used to validate the AI model and evaluation codes.